California State University, Long Beach

Mechanical and Aerospace Engineering Department

MAE 300: Engineering Instrumentation and Measurement

Experiment No. 4

Measurements of Mean Circumferential Pressure Distribution of a Smooth Cylinder in a Cross Flow in Sub-Critical Reynolds Number Regime

Lab Section 19

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Abstract:

The objective of this experiment is to determine the drag coefficient of a cylindrical object by measuring the pressure distribution on the surface of the object. By rotating the cylindrical object in increments of 5 degrees in the clockwise direction until a full rotation is achieved, measurements of the different pressure values will be recorded with a digital manometer. The circumferential pressure difference resulting from the flow separation are calculated. The experiment involves two separate test runs with the speed of the airflow being 15 Hz and then 25 Hz and the same process of incrementally rotating the cylinder will be followed for each of the test runs. After gathering the measurements of the pressures at each of the rotational measurements the pressure and drag coefficients are calculated and graphed to find the flow acceleration, deceleration, and separation, as well as the angles of flow separation. The Reynold's number and the uncertainty of the drag coefficient will also be calculated using data collected in the experiment.

Introduction:

There are many different applications in which an object is exposed to external fluid flow such as a racecar, commercial airplane, or even a golf ball. Each of these different objects are subjected to a multitude of tests to determine its interaction with the surrounding fluid. Understanding the behavior of the flow of a fluid around an object is important as it directly impacts the performance and efficiency of these systems. Understanding the procedure to test and calculate significant values such as the Reynolds number, drag coefficient, and the pressure coefficient (C_p) are critical in understanding how to improve the design of such systems.

The Reynolds number (Re) is a dimensionless value that represents the ratio of the inertial force to the viscous force within an object immersed in a specific fluid. The Reynolds number is key in analyzing the fluid flow conditions, more specifically whether the flow is turbulent or laminar. Within an engineering context, flow that is considered laminar is one with a Re of less than 10^5, while turbulent flow is considered to have a Re of 10^5 or higher. In addition to classifying the type of fluid flow, the Re is significant when determining the drag force around an object as well as the flow separation and stagnation pressure in the wake of an object. Another significant value is the dynamic pressure which is the difference between the total pressure and the static pressure. The dynamic pressure is the pressure in a moving fluid, and it can increase or decrease depending on the fluid motion. To measure the dynamic pressure one device is required to measure the total pressure by facing the direction of fluid flow which provides the total pressure value, while another device is oriented perpendicular to the flow direction to measure the static pressure. The device used to measure the total pressure is the Pitot tube. This device was invented by French engineer Henri Pitot in 1732, and it is a slender cylindrical tube that creates a stagnation point of the fluid flow by decelerating the fluid flow to zero velocity. A manometer is another device that is used to measure static pressure.

Knowledge of these procedures as well as understanding the calculations of the C_p and drag coefficient provides invaluable information regarding the behavior of the external fluid flow

around an object. The drag coefficient is a dimensionless number that represents the impact of shape, orientation, and other pertinent flow conditions of a body exposed to external fluid flow. Knowledge of the flow acceleration, deceleration, separation, as well as flow separation angles inform the improvements design of a multitude of different mechanical systems.

Background & Theory:

The purpose of this experiment is to analyze the flow of air over a smooth circular cylinder and determine how pressure distribution around the cylinder relates to drag. This was accomplished using an open-circuit wind tunnel, where a cylindrical test object was placed in a wind tunnel. Pressure measurements were taken at different angular positions around the cylinder by rotating it in increments of 5° for a total of 73 measurements (covering 0° to 360°). These measurements allowed for a full circumferential pressure distribution to be obtained.

The flow behavior around a cylinder is primarily governed by the Reynolds number, given in Eq. (9). This dimensionless number depends on the fluid density, free-stream velocity, cylinder diameter, and dynamic viscosity. It indicates whether the flow is laminar or turbulent and determines where flow separation occurs. As the Reynolds number increases, flow separation shifts and significantly affects the pressure distribution and resulting drag force.

The first step in the analysis is to determine the pressure coefficient using Eq. (1). This equation normalizes the measured pressure at each angle relative to the free-stream and stagnation pressures, making it a dimensionless quantity. The pressure coefficient describes how pressure varies around the surface of the cylinder and is essential for identifying flow characteristics such as stagnation points and separation regions. In addition, Eq. (2) is used to calculate the change in pressure coefficient, which helps quantify pressure differences around the cylinder.

Once the pressure coefficient distribution is known, it is used to calculate the drag coefficient experimentally using Eq. (3). This equation integrates the pressure coefficient around the entire surface of the cylinder, accounting for the contribution of pressure forces in the flow direction. Essentially, it converts the pressure distribution into a single value that represents the total drag force acting on the cylinder in dimensionless form.

To ensure accurate results, uncertainty analysis is performed using Eqs. (4)–(7). These equations account for measurement errors in pressure readings and propagate them into the calculated drag coefficient. Eq. (7) determines the uncertainty in the pressure coefficient, which is then used in Eqs. (4)–(6) to estimate the uncertainty in the drag coefficient. This step is important for evaluating the reliability of the experimental results.

The free-stream velocity of the air is determined using Eq. (8), which relates velocity to the pressure difference measured at the stagnation point. This velocity is then used in Eq. (9) to calculate the Reynolds number. The Reynolds number allows comparison of the experimental drag coefficient with standard reference data.

Finally, the experimental drag coefficient is compared to the theoretical or reference drag coefficient obtained from empirical charts based on Reynolds number. The difference between

these values is expressed using Eq. (10), and the final experimental value with its uncertainty is reported using Eq. (11). This comparison helps evaluate how closely the experiment matches theoretical expectations and highlights any sources of error.

Overall, each equation plays a specific role in the analysis and is directly connected. The measured pressures are first converted into pressure coefficients (Eqs. (1)–(2)), which are then used to calculate drag (Eq. (3)). Velocity (Eq. (8)) feeds into Reynolds number (Eq. (9)), linking the experiment to theoretical data. Finally, uncertainty equations (Eqs. (4)–(7), (10), & (11)) ensure the results are accurate and reliable. Together, these steps provide a complete understanding of airflow behavior and drag on a circular cylinder.

1) $C_P = \frac{P_\theta - P_\infty}{P_\theta - P_\infty|_{\theta=0}}$
2) $\Delta C_P = \frac{\Delta P}{P_\theta - P_\infty}$
3) $C_d = \frac{F_d}{\frac{1}{2}\rho U_\infty^2 A} = \frac{1}{2}\sum_{\theta-0}^{\theta=2\pi} C_p \cos\theta d\theta$
4) $U_{cd} = \left[\left(\frac{\sum C_p \delta C_d}{C_d \sum C_p} U_{cp}\right)^2 + \left(\frac{f(\theta)\delta C_d}{C_d f(\theta)} U_{f(\theta)}\right)^2\right]^{\frac{1}{2}}$
5) $U_{cd} = \pm\left[\left(\frac{\sum C_p \delta C_d}{C_d \sum C_p} U_{cp}\right)^2\right]^{\frac{1}{2}}$
6) $U_{cd} = \pm\left[\left(U_{\Sigma cp}\right)^2\right]^{\frac{1}{2}}$
7) $U_{\Sigma cp} = \frac{\Sigma\left(\frac{\Delta P}{P_\theta - P_\infty|_{\theta=0}}\right)}{n}$
8) $U_\infty = \sqrt{\frac{2*(P_\theta - P_\infty)|_{\theta=0}}{\rho}}$
9) $Re_D = \frac{\rho U_\infty D}{\mu}$
10) $U_{Cd} = \frac{\Delta C_d}{C_d}$
11) $C_{d\ experimental} = C_d \pm \Delta C_d$

Experimental Procedures:

The experiment began by calibrating the digital manometer by turning it on and waiting a few minutes for the apparatus to warm up. The manometer zeroed and the knob was locked. Students placed the smooth cylinder inside the wind tunnel and connected its pressure tap to the positive pressure side of the manometer. After the cylinder was set inside the wind tunnel, a baseline reading of -0.159 was recorded as a baseline to subtract from all subsequent data. The wind tunnel was then turned on and the ambient pressure (P_∞) was recorded to subtract from all subsequent data. The speed of the wind tunnel was set to the lowest possible setting for the first test run. Next, the circular cylinder was manually rotated in 5° intervals from 0° to 360° to provide 73 pressure measurements. At each interval, a high and low pressure reading was

collected, and the resulting mean pressure differential was calculated at each position. This process was done for 15 Hz and 25 Hz.

Raw data:

Figure 1.1: 15 Hz Data Test

Angle	P	P	P	ΔP	$P_\theta - P_\infty$	C_p	$C_p cos\theta d\theta$	ΔC_p
Degree	max	min	average					
0	-0.090	-0.088	-0.089	0.002	0.068	1.000	0.044	0.029
5	-0.095	-0.090	-0.093	0.005	-0.17977	-2.644	-0.115	0.074
10	-0.098	-0.097	-0.098	0.001	-0.0975	-1.434	-0.062	0.015
15	-0.106	-0.102	-0.104	0.004	-0.104	-1.529	-0.064	0.059
20	-0.116	-0.112	-0.114	0.004	-0.114	-1.676	-0.069	0.059
25	-0.125	-0.123	-0.124	0.002	-0.124	-1.824	-0.072	0.029
30	-0.140	-0.138	-0.139	0.002	-0.139	-2.044	-0.077	0.029
35	-0.153	-0.153	-0.153	0	-0.153	-2.250	-0.080	0.000
40	-0.167	-0.165	-0.166	0.002	-0.166	-2.441	-0.082	0.029
45	-0.180	-0.179	-0.180	0.001	-0.1795	-2.640	-0.081	0.015
50	-0.194	-0.192	-0.193	0.002	-0.193	-2.838	-0.080	0.029
55	-0.206	-0.204	-0.205	0.002	-0.205	-3.015	-0.075	0.029
60	-0.217	-0.214	-0.216	0.003	-0.2155	-3.169	-0.069	0.044
65	-0.224	-0.221	-0.223	0.003	-0.2225	-3.272	-0.060	0.044
70	-0.230	-0.226	-0.228	0.004	-0.228	-3.353	-0.050	0.059
75	-0.238	-0.230	-0.234	0.008	-0.234	-3.441	-0.039	0.118
80	-0.228	-0.222	-0.225	0.006	-0.225	-3.309	-0.025	0.088
85	-0.223	-0.217	-0.220	0.006	-0.22	-3.235	-0.012	0.088
90	-0.217	-0.214	-0.216	0.003	-0.2155	-3.169	0.000	0.044
95	-0.215	-0.212	-0.214	0.003	-0.2135	-3.140	0.012	0.044
100	-0.215	-0.211	-0.213	0.004	-0.213	-3.132	0.024	0.059
105	-0.214	-0.212	-0.213	0.002	-0.213	-3.132	0.035	0.029
110	-0.213	-0.211	-0.212	0.002	-0.212	-3.118	0.047	0.029
115	-0.214	-0.211	-0.213	0.003	-0.2125	-3.125	0.058	0.044
120	-0.215	-0.211	-0.213	0.004	-0.213	-3.132	0.068	0.059
125	-0.215	-0.212	-0.214	0.003	-0.2135	-3.140	0.079	0.044
130	-0.217	-0.214	-0.216	0.003	-0.2155	-3.169	0.089	0.044
135	-0.215	-0.211	-0.213	0.0039	-0.2131	-3.133	0.097	0.057
140	-0.216	-0.214	-0.215	0.002	-0.215	-3.162	0.106	0.029
145	-0.215	-0.212	-0.214	0.003	-0.2135	-3.140	0.112	0.044
150	-0.215	-0.213	-0.214	0.002	-0.214	-3.147	0.119	0.029
155	-0.217	-0.214	-0.216	0.003	-0.2155	-3.169	0.125	0.044
160	-0.217	-0.213	-0.215	0.004	-0.215	-3.162	0.130	0.059
165	-0.219	-0.213	-0.216	0.006	-0.216	-3.176	0.134	0.088
170	-0.218	-0.213	-0.216	0.005	-0.2155	-3.169	0.136	0.074
175	-0.216	-0.212	-0.214	0.004	-0.214	-3.147	0.137	0.059

180	-0.216	-0.212	-0.214	0.004	-0.214	-3.147	0.137	0.059
185	-0.217	-0.210	-0.214	0.007	-0.2135	-3.140	0.136	0.103
190	-0.217	-0.214	-0.216	0.003	-0.2155	-3.169	0.136	0.044
195	-0.217	-0.213	-0.215	0.004	-0.215	-3.162	0.133	0.059
200	-0.217	-0.213	-0.215	0.004	-0.215	-3.162	0.130	0.059
205	-0.215	-0.212	-0.214	0.003	-0.2135	-3.140	0.124	0.044
210	-0.216	-0.211	-0.214	0.005	-0.2135	-3.140	0.119	0.074
215	-0.216	-0.213	-0.215	0.003	-0.2145	-3.154	0.113	0.044
220	-0.217	-0.213	-0.215	0.004	-0.215	-3.162	0.106	0.059
225	-0.217	-0.215	-0.216	0.002	-0.216	-3.176	0.098	0.029
230	-0.216	-0.213	-0.215	0.003	-0.2145	-3.154	0.088	0.044
235	-0.218	-0.216	-0.217	0.002	-0.217	-3.191	0.080	0.029
240	-0.219	-0.216	-0.218	0.003	-0.2175	-3.199	0.070	0.044
245	-0.221	-0.217	-0.219	0.004	-0.219	-3.221	0.059	0.059
250	-0.222	-0.217	-0.220	0.005	-0.2195	-3.228	0.048	0.074
255	-0.223	-0.218	-0.221	0.005	-0.2205	-3.243	0.037	0.074
260	-0.223	-0.219	-0.221	0.004	-0.221	-3.250	0.025	0.059
265	-0.223	-0.219	-0.221	0.004	-0.221	-3.250	0.012	0.059
270	-0.228	-0.225	-0.227	0.003	-0.2265	-3.331	0.000	0.044
275	-0.239	-0.232	-0.236	0.007	-0.2355	-3.463	-0.013	0.103
280	-0.240	-0.234	-0.237	0.006	-0.237	-3.485	-0.026	0.088
285	-0.234	-0.227	-0.231	0.007	-0.2305	-3.390	-0.038	0.103
290	-0.225	-0.220	-0.223	0.005	-0.2225	-3.272	-0.049	0.074
295	-0.214	-0.210	-0.212	0.004	-0.212	-3.118	-0.057	0.059
300	-0.200	-0.198	-0.199	0.002	-0.199	-2.926	-0.064	0.029
305	-0.188	-0.184	-0.186	0.004	-0.186	-2.735	-0.068	0.059
310	-0.172	-0.170	-0.171	0.002	-0.171	-2.515	-0.071	0.029
315	-0.158	-0.155	-0.157	0.003	-0.1565	-2.301	-0.071	0.044
320	-0.143	-0.140	-0.142	0.003	-0.1415	-2.081	-0.070	0.044
325	-0.129	-0.125	-0.127	0.004	-0.127	-1.868	-0.067	0.059
330	-0.120	-0.113	-0.117	0.007	-0.1165	-1.713	-0.065	0.103
335	-0.107	-0.104	-0.106	0.003	-0.1055	-1.551	-0.061	0.044
340	-0.100	-0.095	-0.098	0.005	-0.0975	-1.434	-0.059	0.074
345	-0.098	-0.093	-0.096	0.005	-0.0955	-1.404	-0.059	0.074
350	-0.096	-0.091	-0.094	0.005	-0.0935	-1.375	-0.059	0.074
355	-0.096	-0.089	-0.093	0.007	-0.0925	-1.360	-0.059	0.103
360	-0.096	-0.089	-0.093	0.007	-0.0925	-1.360	-0.059	0.103

Figure 1.2: 25 Hz Data Test

Angle	P	P	P	ΔP	$P_\theta - P_\infty$	C_p	$C_p cos\theta d\theta$	ΔC_p
Degree	max	min	average					
0	0.029	0.055	0.042	0.026	0.199	1.000	0.0436	0.131

5	0.022	0.051	0.037	0.029	0.1935	0.972	0.0423	0.146
10	0.005	0.030	0.018	0.025	0.1745	0.877	0.0377	0.126
15	-0.009	0.010	0.001	0.019	0.1575	0.791	0.0334	0.095
20	-0.021	0.036	0.008	0.057	0.1645	0.827	0.0339	0.286
25	-0.064	-0.074	-0.069	-0.01	0.088	0.442	0.0175	-0.050
30	-0.104	-0.110	-0.107	-0.006	0.05	0.251	0.0095	-0.030
35	-0.148	-0.155	-0.152	-0.007	0.0055	0.028	0.0010	-0.035
40	-0.190	-0.200	-0.195	-0.01	-0.038	-0.191	-0.0064	-0.050
45	-0.236	-0.242	-0.239	-0.006	-0.082	-0.412	-0.0127	-0.030
50	-0.269	-0.384	-0.327	-0.115	-0.1695	-0.852	-0.0239	-0.578
55	-0.307	-0.320	-0.314	-0.013	-0.1565	-0.786	-0.0197	-0.065
60	-0.333	-0.359	-0.346	-0.026	-0.189	-0.950	-0.0207	-0.131
65	-0.348	-0.372	-0.360	-0.024	-0.203	-1.020	-0.0188	-0.121
70	-0.383	-0.395	-0.389	-0.012	-0.232	-1.166	-0.0174	-0.060
75	-0.387	-0.410	-0.399	-0.023	-0.2415	-1.214	-0.0137	-0.116
80	-0.374	-0.399	-0.387	-0.025	-0.2295	-1.153	-0.0087	-0.126
85	-0.362	-0.382	-0.372	-0.02	-0.215	-1.080	-0.0041	-0.101
90	-0.337	-0.360	-0.349	-0.023	-0.1915	-0.962	0.0000	-0.116
95	-0.327	-0.354	-0.341	-0.027	-0.1835	-0.922	0.0035	-0.136
100	-0.330	-0.352	-0.341	-0.022	-0.184	-0.925	0.0070	-0.111
105	-0.327	-0.350	-0.339	-0.023	-0.1815	-0.912	0.0103	-0.116
110	-0.330	-0.349	-0.340	-0.019	-0.1825	-0.917	0.0137	-0.095
115	-0.329	-0.352	-0.341	-0.023	-0.1835	-0.922	0.0170	-0.116
120	-0.338	-0.370	-0.354	-0.032	-0.197	-0.990	0.0216	-0.161
125	-0.342	-0.365	-0.354	-0.023	-0.1965	-0.987	0.0247	-0.116
130	-0.340	-0.357	-0.349	-0.017	-0.1915	-0.962	0.0270	-0.085
135	-0.340	-0.365	-0.353	-0.025	-0.1955	-0.982	0.0303	-0.126
140	-0.339	-0.366	-0.353	-0.027	-0.1955	-0.982	0.0328	-0.136
145	-0.338	-0.366	-0.352	-0.028	-0.195	-0.980	0.0350	-0.141
150	-0.341	-0.369	-0.355	-0.028	-0.198	-0.995	0.0376	-0.141
155	-0.340	-0.359	-0.350	-0.019	-0.1925	-0.967	0.0383	-0.095
160	-0.341	-0.367	-0.354	-0.026	-0.197	-0.990	0.0406	-0.131
165	-0.339	-0.368	-0.354	-0.029	-0.1965	-0.987	0.0416	-0.146
170	-0.333	-0.368	-0.351	-0.035	-0.1935	-0.972	0.0418	-0.176
175	-0.346	-0.362	-0.354	-0.016	-0.197	-0.990	0.0430	-0.080
180	-0.346	-0.362	-0.354	-0.016	-0.197	-0.990	0.0432	-0.080
185	-0.346	-0.362	-0.344	-0.016	-0.187	-0.940	0.0408	-0.080
190	-0.333	-0.368	-0.348	-0.035	-0.191	-0.960	0.0412	-0.176
195	-0.339	-0.368	-0.345	-0.029	-0.188	-0.945	0.0398	-0.146
200	-0.341	-0.367	-0.343	-0.026	-0.186	-0.935	0.0383	-0.131
205	-0.340	-0.359	-0.342	-0.019	-0.185	-0.930	0.0368	-0.095
210	-0.341	-0.369	-0.336	-0.028	-0.179	-0.899	0.0340	-0.141
215	-0.338	-0.366	-0.347	-0.028	-0.19	-0.955	0.0341	-0.141

220	-0.339	-0.366	-0.344	-0.027	-0.187	-0.940	0.0314	-0.136
225	-0.340	-0.365	-0.336	-0.025	-0.179	-0.899	0.0278	-0.126
230	-0.340	-0.357	-0.338	-0.017	-0.181	-0.910	0.0255	-0.085
235	-0.342	-0.365	-0.336	-0.023	-0.179	-0.899	0.0225	-0.116
240	-0.338	-0.370	-0.340	-0.032	-0.183	-0.920	0.0201	-0.161
245	-0.329	-0.352	-0.337	-0.023	-0.18	-0.905	0.0167	-0.116
250	-0.330	-0.349	-0.340	-0.019	-0.183	-0.920	0.0137	-0.095
255	-0.327	-0.350	-0.340	-0.023	-0.183	-0.920	0.0104	-0.116
260	-0.330	-0.352	-0.350	-0.022	-0.193	-0.970	0.0073	-0.111
265	-0.327	-0.354	-0.360	-0.027	-0.203	-1.020	0.0039	-0.136
270	-0.337	-0.360	-0.368	-0.023	-0.211	-1.060	0.0000	-0.116
275	-0.362	-0.382	-0.392	-0.02	-0.235	-1.181	-0.0045	-0.101
280	-0.374	-0.399	-0.385	-0.025	-0.228	-1.146	-0.0087	-0.126
285	-0.387	-0.410	-0.367	-0.023	-0.21	-1.055	-0.0119	-0.116
290	-0.383	-0.395	-0.347	-0.012	-0.19	-0.955	-0.0142	-0.060
295	-0.348	-0.372	-0.318	-0.024	-0.161	-0.809	-0.0149	-0.121
300	-0.333	-0.359	-0.285	-0.026	-0.128	-0.643	-0.0140	-0.131
305	-0.307	-0.320	-0.244	-0.013	-0.087	-0.437	-0.0109	-0.065
310	-0.269	-0.384	-0.201	-0.115	-0.044	-0.221	-0.0062	-0.578
315	-0.236	-0.242	-0.154	-0.006	0.003	0.015	0.0005	-0.030
320	-0.190	-0.200	-0.110	-0.01	0.047	0.236	0.0079	-0.050
325	-0.148	-0.155	-0.071	-0.007	0.086	0.432	0.0154	-0.035
330	-0.104	-0.110	-0.033	-0.006	0.124	0.623	0.0235	-0.030
335	-0.064	-0.074	-0.008	-0.01	0.149	0.749	0.0296	-0.050
340	-0.021	0.036	0.020	0.057	0.177	0.889	0.0365	0.286
345	-0.009	0.010	0.029	0.019	0.186	0.935	0.0394	0.095
350	0.005	0.030	0.030	0.025	0.187	0.940	0.0404	0.126
355	0.022	0.051	0.021	0.029	0.178	0.894	0.0389	0.146
360	0.029	0.055	0.034	0.026	0.191	0.960	0.0419	0.131

Sample Calculations and Results:

Test Run 1 (15 Hz) Sample Calculations:

Pressure Coefficient (C_P) at $\theta = 15°$:

$$C_P = \frac{P_\theta - P_\infty}{P_\theta - P_\infty|_{\theta=0}} = \frac{0.053}{0.068} = 0.779$$

Experimental Drag Coefficient (C_d) for 15 Hz Test Run:

$$d\theta = 5° = 0.0873 \text{ rad}$$

$$C_d = \frac{1}{2}\sum_{\theta-0}^{\theta=2\pi} C_p \cos\theta d\theta = \frac{1}{2}(2.256) = 1.128$$

Fluid Velocity (U_∞) for 15 Hz Test Run:

Assume that room temperature is 23 °C: $\rho = 1.194\ kg/m^3$

$1\ torr = 133.32\ Pa$

$$U_\infty = \sqrt{\frac{2 * (\mathrm{P_\theta} - \mathrm{P_\infty})|_{\theta=0}}{\rho}} = \sqrt{\frac{2 * (0.068)(133.32)}{1.194}} = 3.90\ m/s$$

Reynold's Number for 15 Hz Test Run:

$D = 0.00635\ m$

$\mu = 1.826E - 05\ kg/ms$

$$Re_D = \frac{\rho U_\infty D}{\mu} = \frac{(1.194)(3.897)(0.00635)}{1.826E - 05} = 1618$$

Uncertainty of Experimental $\mathrm{C_d}$:

$\Sigma(\Delta C_P) = 4.043$

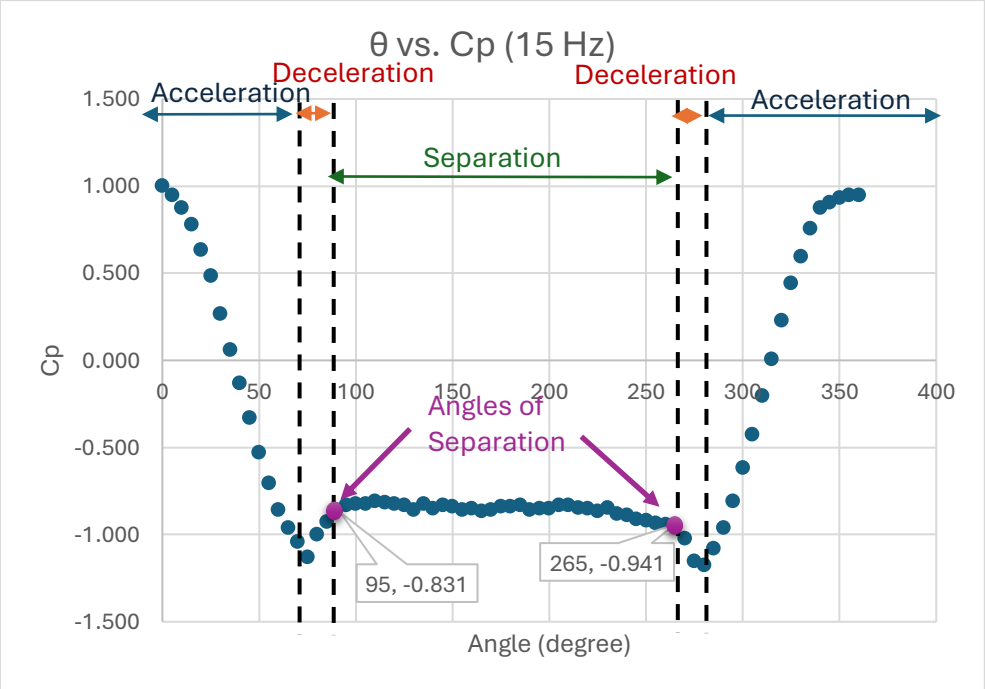
$$U_{\Sigma cp} = \frac{\Sigma\left(\frac{\Delta P}{\mathrm{P_\theta} - \mathrm{P_\infty}|_{\theta=0}}\right)}{n} = \frac{\Sigma(\Delta C_P)}{n} = \frac{4.043}{73} = 0.055$$

$$U_{cd} = \pm\left[\left(U_{\Sigma cp}\right)^2\right]^{\frac{1}{2}} = \pm[(0.055)^2]^{\frac{1}{2}} = \pm 0.055$$

$$\Delta C_d = U_{cd} * \mathrm{C_d} = 0.055 * 1.128 = 0.062$$

$$C_{d\ experimental} = C_d \pm \Delta C_d = 1.128 \pm 0.062$$

Figure 1.3: Angle versus C_p for 15 Hz Test Run



Test Run 1 (25 Hz) Sample Calculations:

Pressure Coefficient (C_P) at $\theta = 15°$:

$$C_P = \frac{P_\theta - P_\infty}{P_\theta - P_\infty|_{\theta=0}} = \frac{0.199}{0.158} = 0.791$$

Pressure Coefficient (C_P) for 25 Hz Test Run:

$d\theta = 5° = 0.0873 \text{ rad}$

$$C_d = \frac{1}{2}\sum_{\theta-0}^{\theta=2\pi} C_p \cos\theta d\theta = \frac{1}{2}(2.429) = 1.215$$

Fluid Velocity (U_∞) for 25 Hz Test Run:

Assume that room temperature is 23 °C: $\rho = 1.194\ kg/m^3$

$1\ torr = 133.32\ Pa$

$$U_\infty = \sqrt{\frac{2 * (P_\theta - P_\infty)|_{\theta=0}}{\rho}} = \sqrt{\frac{2 * (0.199)(133.32)}{1.194}} = 6.67\ m/s$$

Reynold's Number for 25 Hz Test Run:

$D = 0.00635\ m$

$\mu = 1.826E-05\ kg/ms$

$$Re_D = \frac{\rho U_\infty D}{\mu} = \frac{(1.194)(6.67)(0.00635)}{1.826E-05} = 2768$$

Uncertainty of Experimental C_d:

$\Sigma(\Delta C_P) = -5.9397$

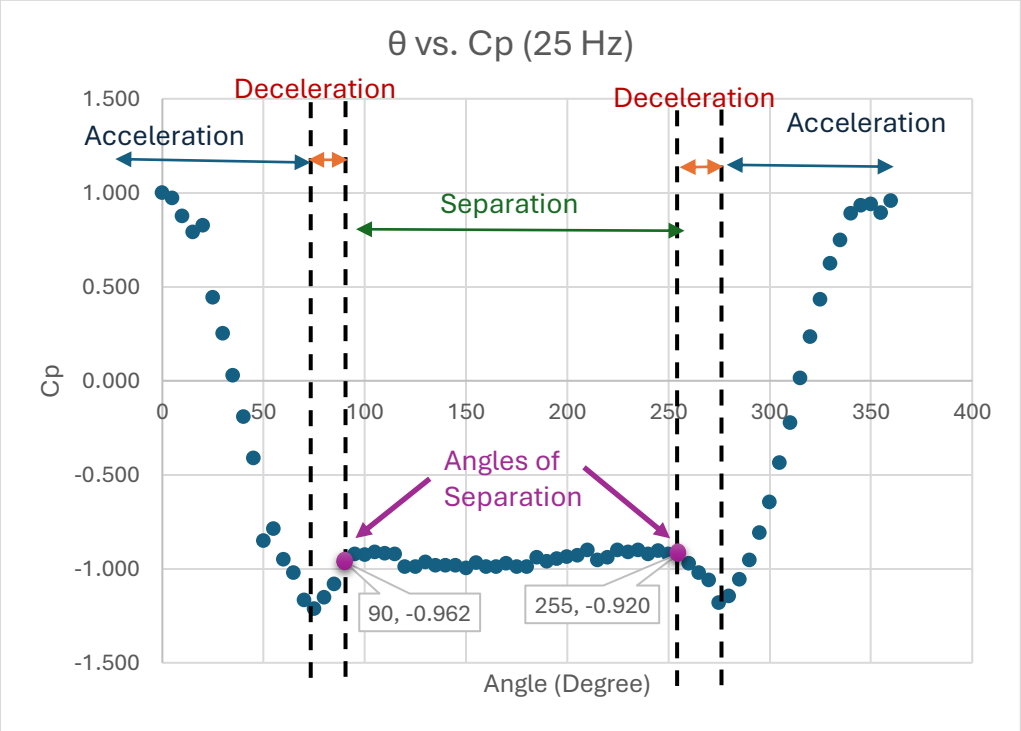
$$U_{\Sigma cp} = \frac{\Sigma\left(\frac{\Delta P}{P_\theta - P_\infty|_{\theta=0}}\right)}{n} = \frac{\Sigma(\Delta C_P)}{n} = \frac{-5.9397}{73} = -0.0814$$

$$U_{cd} = \pm\left[\left(U_{\Sigma cp}\right)^2\right]^{\frac{1}{2}} = \pm[(-0.0814)^2]^{\frac{1}{2}} = \pm 0.0814$$

$$\Delta C_d = U_{cd} * C_d = 0.0814 * 1.215 = 0.0988$$

$$C_{d\ experimental} = C_d \pm \Delta C_d = 1.215 \pm 0.0988$$

Figure 1.4: Angle versus C_p for 25 Hz Test Run



Discussion of results:

An analysis of the θ vs. C_p graph provides insight into three significant regions of interest in the test run cycles. These regions include flow acceleration, flow deceleration, and flow separation as well as the specific angles of flow separation. The region of flow acceleration is the area where the velocity of the fluid flow in the wind tunnel accelerates and the pressure decreases. The region of flow deceleration is where the velocity decelerates and the pressure increases. The pressure and velocity are inversely proportional because according to Bernoulli's equation for incompressible flow ($P + \frac{1}{2}\rho V^2 = constant$) any decrease in velocity must be accompanied by an increase in pressure to ensure a constant sum. The region of flow separation is a region in which the pressure is nearly constant aligning with the nearly constant string of values shown in Figures 1.3-4. An observation of both the 15 Hz and 25 Hz graphs showcase the similarity in the locations of the different regions of interest. For instance, in both the 15 Hz and 25 Hz graph, the first region of flow deceleration start at an angle of 75°, while they both end around $90° - 95°$. The second region of deceleration has slightly more variation in range as the 15 Hz region of deceleration occurs between $265° < \theta < 280°$ while the 25 Hz region occurs between $255° < \theta < 275°$.

The range of angles for which the flow separation occurs is very similar for both test runs as the 15 Hz's region of separation occurs between $95° \leq \theta \leq 265°$ and the 25 Hz region occurs between $90° \leq \theta \leq 255°$. This shows that at an angle of around $92.5°$ the region of flow separation begins and at an angle of around $260°$ the separation stops. Because the flow separation regions occurs between these two angles, the flow around a vertical cylindrical object would begin to separate around the right and left side of the hollow cylinder. An observation aligning with diagrams and other visual representation of fluid streamlines around circular object as shown in Figure 1.5. The region of separation occurs across a span of around $160°$, which is most of the non-frontal face of the object, meaning that the posterior side of the cylinder experiences a region of low pressure. In contrast, the regions of acceleration experience high pressure. The two regions of acceleration for the 15 Hz run occur between $0° < \theta < 75°$ and $265° < \theta < 360°$ and occur between $0° < \theta < 75°$ and $255° < \theta < 360°$ for the 25 Hz run. The starting and ending value for C_P for both graphs is 1, which is consistent with the fact that at angle $0°$ and $360°$ the small port trap on the hollow cylinder is directly aligned with the flow from the wind tunnel. When the fluid hits the port trap, its velocity decelerates to zero, creating a stagnation point which has a C_p of 1.

A comparison between the values of the experimental drag coefficient (C_d) and the theoretical drag coefficient facilitates the validation of the experimental data. The value for the experimental C_d for 15 Hz is 1.128 ± 0.062 and 1.215 ± 0.0988 for 25 Hz. As there was an error with the wind tunnel during the collection of the experimental data, the data used for this analysis was provided by the instructor. However, there were some issues when finding the theoretical value for C_d as the Reynolds number (Re) for both the 15 Hz and 25 Hz test runs were both below four thousand, 1618 for 15 Hz and 2768 for 25 Hz. As the provided a chart (Figure 1.6) to find the theoretical C_d only provided values of C_d for $Re > 4 \times 10^3$ the calculated Re values for the data were unable to find an accurate theoretical C_d. However, looking at the chart in Figure 1.7 the theoretical C_d value for a circular cylinder is 1.17. The percentage error for the 15 Hz C_d value is around 3.6% and 3.8% for the 25 Hz, which means that the experimental values for the C_d are quite similar to the theoretical value from Figure 1.7.

Figure 1.5: Fluid Flow Around Different Shapes

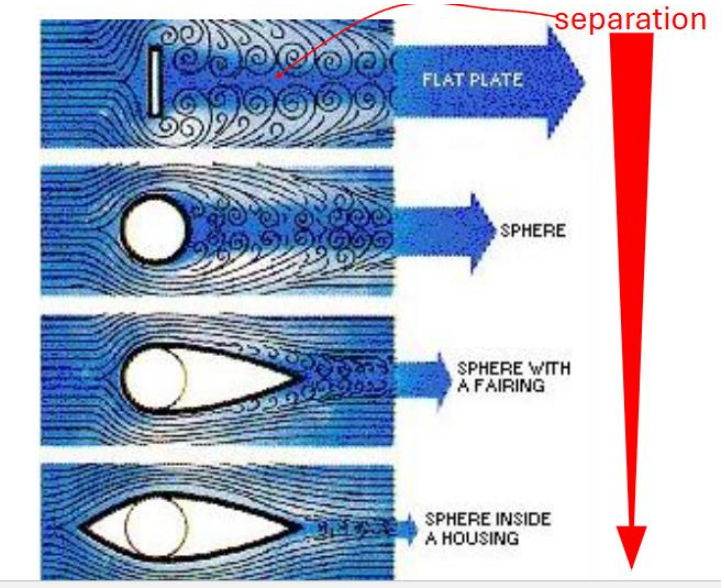
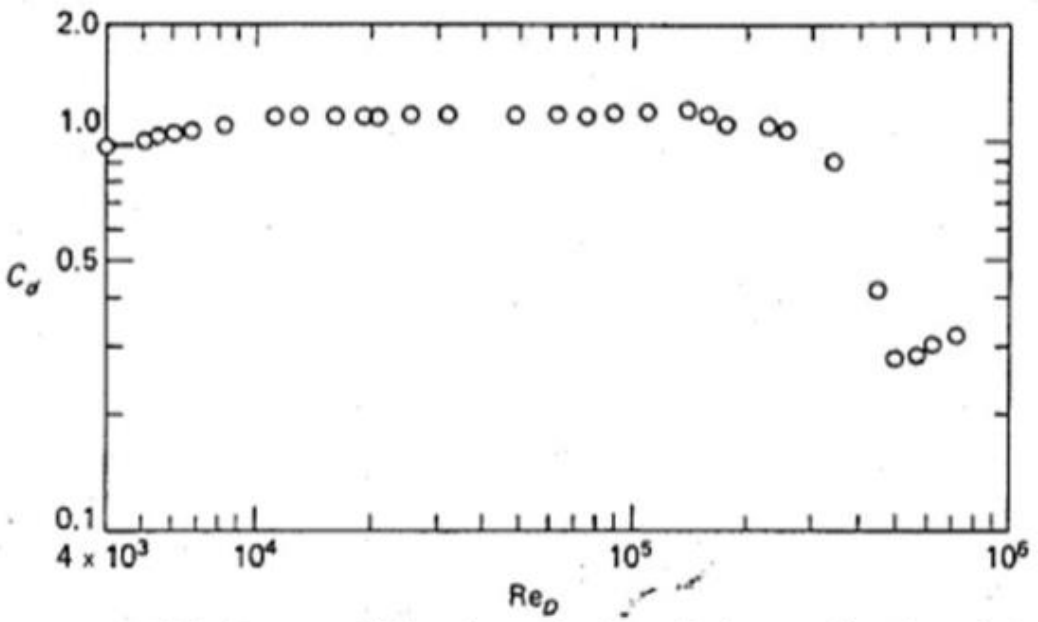
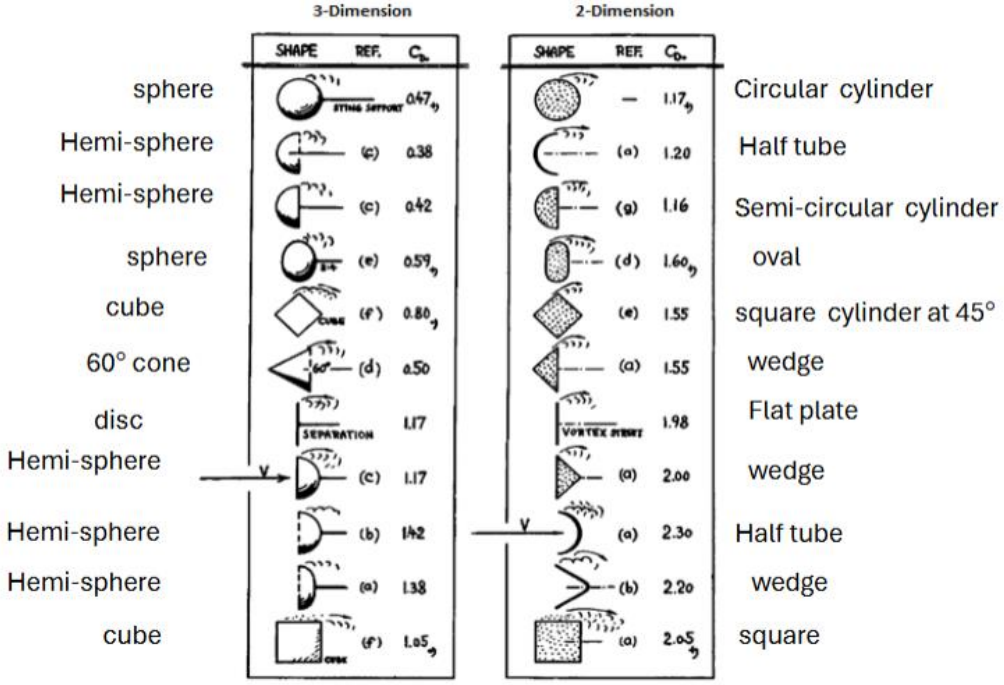


Figure 1.6: Drag Coefficient for Circular Cylinder as a function of Reynold Number



Drag coefficient for a circular cylinder as a function of the Reynolds number. From Boundary Layer Theory *by Schlichting (1968).*

Figure 1.7: Values of Drag Coefficient for Various 2D and 3D Shapes



Conclusions and Recommendations:

The values of the experimental C_d, C_p, Re, and fluid velocity were calculated from the incremental rotation of a hollow cylinder in a wind tunnel running at 15 Hz and 25 Hz. The value of the experimental drag coefficient for the 15 Hz is 1.128 ± 0.062 and 1.215 ± 0.0988 for 25 Hz. The C_p for both wind tunnel speeds are plotted on graphs shown in Figure 1.3-4. The calculated Reynolds number for 15 Hz is 1618 and 2768 for 25 Hz and the fluid velocity is 3.90 m/s for 15 Hz and 6.67 m/s for 25 Hz.

As the values for the Reynolds numbers were both lower than four thousand the graph show in Figure 1.6 was unable to be used to find the value for the theoretical C_d. However, the chart showing various values of C_d for various shapes (Figure 1.7) was used instead, with a value of 1.17 for a circular cylinder. When comparing the value of the experimental C_d for the 15 Hz test run, there was a percentage error of 3.6% and a percentage error of 3.8% for the 25 Hz run, which provides validity to the collected experimental data.

In addition, an analysis of the graphs of the $\theta \; vs. \; C_p$ for the 15 Hz and 25 Hz provide valuable insight into the behavior of the flow streamlines around the cylinder. The 15 Hz's region of separation occurs between $95° \leq \theta \leq 265°$ and the 25 Hz region occurs between $90° \leq \theta \leq 255°$. This means that within these ranges of angles, the boundary layer detaches from the object's surface and flow separation occurs. These angles are consistent with the image of a flow streamline around a circular object as shown in Figure 1.5 as the point at which the

flow streamlines begin to move away from the circular object's surface are around the top and bottom, which align with the angle of $90°$ and $260°$.

One of the main sources of error would be in the estimation of the theoretical C_d as only one value was provided for the shape that was used in the experiment. In addition, as there was an equipment error during the execution of the experiment, the data that was used in this analysis was not collected directly by the students involved in the report. Meaning that any error in the experimental procedure would be unknown for this data set. However, one suggestion to improve the accuracy of the values would be to decrease the incremental angle from 5°. All in all, the process of recording the data from the wind tunnel and calculating the experimental values is invaluable in understanding the importance of values such as the C_d, C_p, and Re in a variety of engineering applications involving external fluid flow.

Bibliography and References:

Wang, Tom. *Complete Lectures and Experiments File*. California State University Long Beach. https://csulb.instructure.com/courses/116603/files/28679338/download?download_frd=1

Rahai, Hamid R. *Instrumentation and Measurements*. California State University Long Beach. https://csulb.instructure.com/courses/116603/files/28679316/download?download_frd=1